

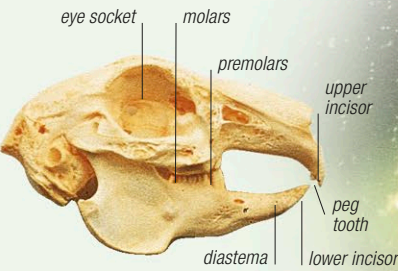
Rabbits, hares, and pikas

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Lagomorpha
FAMILIES	2
SPECIES	92

Although these small to medium-sized gnawing animals are similar to rodents in many ways (for example, both groups have large incisors), they differ in several respects, including the presence of a second set of upper incisors and a lighter skull structure. Rabbits, hares, and pikas (also termed “calling hares” on account of their varied vocalizations)—the lagomorphs—are among the most hunted of all animals. Their natural predators are carnivores and birds, and they are also hunted by humans for sport, for food, and for their fur. All species are terrestrial and are found worldwide (except the West Indies, the southern parts of South America, Madagascar, and several islands of Southeast Asia) in habitats ranging from arctic tundra to semidesert.

Anatomy

The physical features of rabbits and hares reflect their need to perceive danger and elude predators. Large ears provide excellent hearing, eyes positioned high on each side of the head give almost 360-degree vision, and elongated back legs enable impressive running speeds—hares can reach up to 35 mph (56 kph). Unlike rodents, rabbits and hares have tails that are small and spherical, and they have well-furred feet with hair on the soles, which provides grip when running. Pikas tend to hide in crevices or burrows when threatened, and are more mouselike in appearance, having legs of approximately the same length (they cannot run as swiftly). They also have shorter, rounded ears and no visible tail. All species have slitlike nostrils that can be closed completely. Unusual for mammals, the females of some species are larger than the males.



RABBIT SKULL
Lagomorphs have well-developed, continually growing incisor teeth. Behind the upper incisors is a second pair of smaller incisors (“peg teeth”). There is a large gap between the incisors and premolars in both jaws, called the diastema.



FLEXIBLE NECK
In the lagomorphs, self-grooming is important: mutual grooming between individuals is rare. Great flexibility in the neck—this European rabbit is able to rotate its head through 180 degrees—allows them to reach the fur on the back.

Feeding

Lagomorphs are herbivorous and generally eat grass and other succulent plants. Matter that cannot be digested initially is expelled in the form of a moist pellet and eaten, usually straight from the anus. It is then held in the stomach to be mixed with other food for second digestion before being excreted as a dry pellet. In this way, most food travels through the digestive system twice, enabling the animal to derive maximum nutrition from its diet. This process is called refection.



PLANT-EATERS
All lagomorphs, such as this North American pika, spend much of their time feeding. During the summer, the pika also gathers and stores food for winter, creating a “haystack” of dried foliage outside its rocky shelter.

AERIAL BOXING
During the mating season, fights between hares occur frequently. The front feet are used for boxing, and the powerful back feet are used for kicking. Males battle for access to females, and a female will drive off a male if she is not ready to mate (as shown here).



Reproduction

Although lagomorphs are hunted intensely by many predators, they are able to maintain healthy population levels through a high reproductive rate. Because ovulation is not cyclical but is instead triggered in response to copulation, females can become pregnant directly after giving birth. Rabbits, the most prolific breeders of all the lagomorphs, can produce litters of up to 12 young as many as 6 times a year. Furthermore, rabbits are sexually mature at a young age (the European rabbit is able to conceive when only 3 months old), and the gestation period may be very short (the Florida cottontail rabbit, for example, gestates for as little as 26 days).



FEEDING TIME
Although some rabbits give birth in their burrows, all hares are born above ground. These young brown hares remain hidden during the day but are collected together at sunset when the mother visits to nurse them.